Loay Mahmoud

Software Engineer | Product · Full-Stack · Cloud · AWS Certified

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Summary

Product-minded software engineer with **5 years** of experience at startups and scale-ups in the US, Germany, and Egypt. I work directly with clients, founders, and users to turn feedback into shipped features. Former university TA; now a CS Ph.D. student at IU Indianapolis.

Experience

Research Assistant (Part-time) Aug 2026 – Present
Indiana University Indianapolis – CyberWater *Indiana, USA (On-Site)*

- Work directly with CyberWater's users (researchers and Ph.D. students at IU & Pitt), turning feedback into prioritized fixes.
- Refactor and modernize the CyberWater server, APIs, and auto-update system for reliability, security, and maintainability.

Sr. Software Engineer (Full-time) Sep 2025 – Aug 2026
MNT-Halan *Giza, Egypt (On-Site)*

- Built and scaled FinTech-grade ERP modules (accounting, payments, reporting) with Python/Odoo, OWL/JS, and Elixir/Phoenix.
- Gathered requirements from Odoo implementation partners and the accounting team, turning finance workflows into features.
- Led a full Odoo migration that made core financial workflows **5x faster**; prototyped MCP servers for internal LLM tools.

Teaching Assistant (Part-time) Sep 2022 – June 2026
Cairo University *Giza, Egypt (On-Site)*

- Taught tutorials and labs for 3 years: Software Engineering, OOP, Compilers, Advanced Databases, Modeling, and Algorithms.
- Created new labs, exercises, and quizzes, and mentored student teams through semester-long software projects.

Software Engineer II (Full-time) Sep 2024 – Sep 2025
VCloud LLC *Nevada, USA (Remote)*

- Product engineer for **VOCCS**, an omnichannel contact-center platform used by **hundreds of agents** at **a global pharma client**.
- Led weekly client meetings to gather feedback and demo features; demoed to prospective clients' sales and technical teams.
- Replaced static image call scripts with **dynamic agent scripts** after agents reported the images were misleading and overwhelming.
- Refactored backends into modular services on Twilio/AWS Connect (TaskRouter call routing); set up CI/CD on GCP/AWS.

R&D Backend Engineer (Contract) Sep 2024 – Oct 2024
Turing *USA (Remote)*

- Contributed to the SFT Code Interpreter team, training LLMs and AI agents for a MAANG company.

Data Engineer (Part-time) May 2024 – Aug 2024
Pleny *Dubai, UAE (Remote)*

- Unblocked the app launch by fixing data bugs and refactoring validation jobs (Nest.js, MongoDB/Mongoose).

Software Engineer Mar 2022 – Aug 2024
Keepoala *Munich, Germany (Remote)*

- Worked directly with the CTO and CFO (product owner) to build a returns dashboard and loyalty app for online shoppers.
- Helped **nearly triple the user base** by shipping features end-to-end across Node.js backends and React/Next.js frontends.
- Built Shopify and Billbee integrations and set up Sendinblue email marketing to drive shopper engagement.
- Migrated Firestore to PostgreSQL (Prisma); automated deployments with AWS (Docker, ECS) and CI/CD on all repositories.
- Added a backend testing framework (Mocha, Sinon) and Redis caching for the shop plugins (TypeScript).

Software Engineer (Internships & Freelance) Aug 2020 – Sep 2022
_VOIS (Vodafone), Ministry of Water Resources & Irrigation, Master Micro, NajahNow *Egypt*

- Built the Accounts Management feature of the My Vodafone app with the Android core team at _VOIS (Kotlin, Agile/Scrum).
- Sole developer of a Flutter/Firebase field app for the Ministry's site engineers; moved Master Micro's CI to Jenkins.

Education

Indiana University Indianapolis Aug 2026 – Present
Ph.D. in Computer Science. Awarded an Indiana University Fellowship. *Indiana, USA*

Cairo University Sep 2017 – June 2022
B.Sc. Computer and Communication Engineering. Graduated with Honors, GPA 3.65/4.0, ranked 15th in class. *Giza, Egypt*

Skills

Product: Client discovery & demos, requirements gathering, feature scoping, Agile/Scrum, Jira
Languages & Frameworks: TypeScript, Python, Elixir, PHP · React/Next.js, Node.js, Nest.js, Phoenix, Odoo, Laravel, FastAPI
Data & Cloud: PostgreSQL, MySQL, MongoDB, Redis · AWS (ECS, S3, RDS, Connect), GCP, Docker, GitHub Actions
Integrations & AI: Twilio (TaskRouter), Shopify, Billbee, Sendinblue · MCP, LLM training data (SFT)

Certifications & Awards

- AWS Certified Developer – Associate (2024) · 5th place, Dell Envision Future Competition · FIRST LEGO League: 1st in Egypt, Top 25 worldwide